

Civil Jets and Turboprops

The use of the turbine engine for airliners revolutionized air travel. Not only were journey times slashed by more than 50 percent, but the ability to fly above most of the bad weather and the much quieter cabins made traveling a considerably more pleasant experience. Furthermore, as the turbine was more reliable than the hugely complex giant radial engines that powered the last of the piston airliners, dispatch rates improved exponentially.

▷ de Havilland DH106 Comet 1 1952

Origin UK

Engine 4 x 5,000 lb (2,268 kg) thrust de Havilland Ghost turbojet

Top speed 460 mph (740 km/h)

The first production jetliner, the Comet caused a sensation when it entered service. Flying much faster than the propellers with significant flaws in both design and construction resulted in several fatal crashes, causing a loss of confidence in the aircraft.



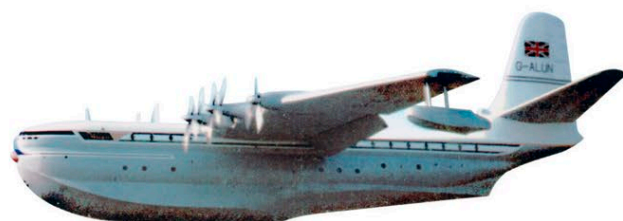
△ Sud Aviation Caravelle 1 1955

Origin France

Engine 2 x 11,400 lb (5,171 kg) thrust Rolls-Royce Avon Mk527 turbojet

Top speed 500 mph (805 km/h)

This first French jetliner was also the first to have engines mounted on the rear of the fuselage. Popular with pilots and passengers, the Caravelle entered service in 1959, the last few retiring as recently as 2004.



△ Saunders Roe SR45 Princess 1952

Origin UK

Engine 10 x 2,250 hp Bristol Proteus 600 turboprop

Top speed 380 mph (610 km/h)

Obsolete before it ever flew, the designers of the Princess failed to take into account several significant factors. These included the number of runways built during WWII, the improved performance of land-based aircraft, and the problems caused by saltwater corrosion.



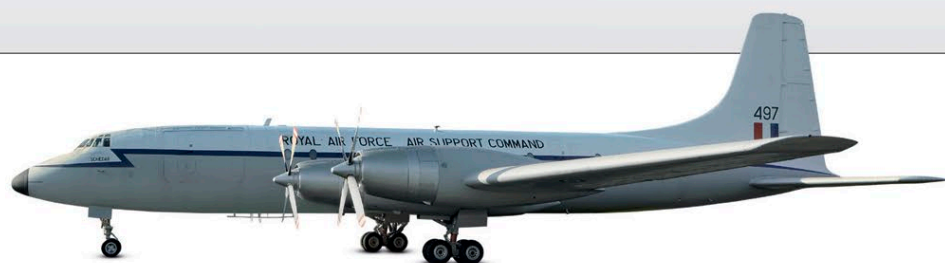
△ Lockheed L188 Electra 1957

Origin US

Engine 4 x 3,750 hp Allison 501-D13 turboprop

Top speed 448 mph (721 km/h)

The Constellation had earned Lockheed's commercial aircraft division a fine reputation, and initially the Electra (which was also the first turboprop airliner produced in the US) sold well. However, a fatal design flaw caused two early crashes and only 170 were built.



△ Bristol Britannia 312 1952

Origin UK

Engine 4 x 4,450 hp Bristol Proteus 765 turboprop

Top speed 397 mph (639 km/h)

Known as the "Whispering Giant," when the Britannia made its maiden flight, it offered a significant increase in performance over the other airliners of the time. Unfortunately, only 85 were built.



△ Tupolev Tu-104 1955

Origin USSR

Engine 2 x 21,400 lb (9,707 kg) thrust Mikulin AM-3M-500 turbojet

Top speed 497 mph (800 km/h)

This was the USSR's first jetliner. Its appearance in London in 1956 carrying Nikita Khrushchev on a State visit caused a considerable stir among Western observers, who were unaware the Soviet aviation industry was so advanced.

▷ Boeing 707 1958

Origin US

Engine 4 x 17,000 lb (7,711 kg) thrust Pratt & Whitney JT3-D turbofan

Top speed 621 mph (1,000 km/h)

One of the most important aircraft of all time, the 707 was the West's first successful jetliner—shown here is the 367-80 prototype for the 707. It was the civil development of the 367-80 ("Dash 80") prototype. More than 1,000 would be built in a number of different versions.

